

APPROVED METHOD:
**GUIDE FOR THE SELECTION, CARE,
AND USE OF ELECTRICAL INSTRUMENTS
IN THE PHOTOMETRIC LABORATORY**
AN AMERICAN NATIONAL STANDARD

ANSI/IES LM-28-20

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Publication of this document
has been approved by the IES.
Suggestions for revision
should be directed to the IES.

**Prepared by
The IES Testing Procedures Committee**



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CONTENTS

1.0	Introduction and Scope	1
1.1	Introduction	1
1.2	Scope	1
2.0	Normative References	1
2.1	ANSI/IES LS-1-20	1
3.0	Definitions and Nomenclature	1
4.0	General Instrument Considerations	1
4.1	Digital vs. Analog Instruments	1
4.2	Single-Range vs. Multi-range Instruments	1
4.3	Calibration	2
4.4	Tolerance Interval	2
5.0	Ambient and Physical Condition Measurements	2
5.1	Temperature Measurements	2
5.2	Air Movement Measurements	2
5.3	Angular Position Sensing	2
6.0	Electrical Measurements	3
6.1	General	3
6.2	Impedance Limitations	3
7.0	Optical Radiation Detectors and Measuring Instruments	4
7.1	General	4
7.2	Optical Radiation Detectors	4
7.3	Amplifiers and Logging/Display Devices	4
7.4	Electrical Filtering	4
7.5	Linearity of Optical Radiation Measurement	5
7.6	Cosine Response	5
7.7	Photopic Response	5

8.0 Computer Interfacing of Photometric Equipment..... 6

9.0 Safety..... 6

9.1 General 6

9.2 Electrical Shock 6

10.0 Additional Resources..... 6

Annex A – Analog Instrument Selection and Considerations 6

Annex B – Tolerance Interval vs. Acceptance Interval. 8

Annex C – Measurement Errors Due to the Deviation in the System Response from $V(\lambda)$ 9

Informative References 11

1.0 Introduction and Scope

1.1 Introduction

ANSI/IES LM-28-20, is a revision of the 2012 report of the same name. References were updated along with the sequence of information.

1.2 Scope

This guide describes terms used to define instrument capabilities for electrical quantities, angular positioning, optical radiation detectors, airflow, and temperature. In addition, the reader is informed about cautions that needs to be observed when selecting, maintaining, and using measuring instruments in the photometric laboratory while maintaining appropriate levels of uncertainty.

2.0 Normative References

2.1 ANSI/IES LS-1-20

Illuminating Engineering Society. ANSI/IES LS-1-20, Lighting Science: Nomenclature and Definitions for Illuminating Engineering. New York: IES; 2020. Online: www.ies.org/standards/definitions/. (Accessed 2019 May 15).

3.0 Definitions and Nomenclature

The units of electrical measurement used in this test method are the volt, the ampere, and the watt.

The term *instrument* will be used throughout the guide instead of the term *meter*, which was incorrectly applied in previous publications of this document.

The instruments required for the photometric laboratory can be divided into two groups: 1) instruments for the measurement of electrical quantities in the testing of lamps and auxiliaries, and 2) special instruments and devices for such measurements as detector output, temperature, source color, total luminous flux, and

luminaire and photometer angular position. Indicating and recording instruments may be used in either type of application.

4.0 General Instrument Considerations

4.1 Digital vs. Analog Instruments

Digital instruments are widely used in the lighting industry. Digital instruments are often preferred over analog instruments because of their accuracy, precision, cost effectiveness, ease of use, and superior stability. High quality analog instruments can also be viable choices for the measurement of voltage, current, power and temperature when properly used. **Annex A** provides information regarding the selection and precautions involved with the use of analog instruments.

4.2 Single-Range vs. Multi-range Instruments

Instruments may be single-range or multi-range. Most digital instruments that are currently available are of the multi-range type. If appropriately used, single-range instruments are equally viable tools in the photometric laboratory.

Most multi-range digital instruments are designed to automatically select the most appropriate range for the values being measured. Regardless of the type of instrument being used, it is critical that care be taken to ensure that the readings are in the upper portion of the scale of the instrument. To highlight the necessity of proper range selection, the following example is provided:

- Given a DC ammeter with a 10-ampere (amp, A) range specified by the manufacturer to be accurate within 0.5% of full scale. Then: 0.5% of 10 amps is 0.05 amp.
- If the measured current is read as 5 A, then the accuracy of 0.05 A represents 1% of the measured value.
- If the measured current is read as 1 A, then the accuracy of 0.05 A represents 5% of the measured value.

4.3 Calibration

Instruments used in the photometric laboratory shall be maintained according to an appropriate calibration schedule. The manufacturer's recommended calibration schedule should be followed unless the laboratory's assessment of the instrument's stability and the required uncertainty for the measurement indicates a revised schedule.

4.4 Tolerance Interval

A tolerance interval is an interval of permissible values of a property. The acceptable results of a measurement lie within an acceptance interval, defined as the tolerance interval reduced by the expanded uncertainty (95% confidence) of the measurement on both limits of the tolerance interval. (See **Annex B** in this document, and ISO/IEC Guide 98-4, Section 3.3.5.)

5.0 Ambient and Physical Condition Measurements

5.1 Temperature Measurements

Room ambient temperature may be measured using thermocouples or resistance temperature detectors (RTD's). A specially designed air temperature probe should be used to maintain uniformity in measurements. If an air temperature probe is not available, a standard thermocouple junction or RTD sensing point should be immersed in 15 mm (0.59 in.) of mineral oil in a glass container. The sensing point of the instrument shall be located at the same height as the test luminaire or lamp. The sensing point should also be located as close as possible to the test luminaire or lamp without allowing the convection currents produced by the lamps to influence the temperature reading. The sensing point shall be shielded from direct radiation from all light sources.

Temperature measurements on luminaire housings or other large areas can be conveniently determined using color changing lacquers, melting waxes, or liquid crystal sheets. Spot temperatures are best measured using infrared pyrometers, thermocouples or RTDs.

5.2 Air Movement Measurements

Drafts can have a significant effect on the output of electrical discharge lamps such as fluorescent and low-pressure sodium lamps. It is therefore important that the heating and cooling systems used within the laboratory are designed with draft reduction in mind. Airflow within the laboratory may be detected and analyzed by means of a hot-wire anemometer, or other sensitive air movement instruments. When measuring draft conditions, the luminaire (or lamp) should be at ambient temperature. Measurements should be taken within 0.6 m (2 ft) of the luminaire (or lamp). Heating and cooling systems should be operating at the time of measurement to observe their effect on drafts. The allowable airflow within the laboratory depends on the sensitivity of the luminaire or lamp that is being measured. (For the allowable airflow limitations for a type lamp or luminaire, consult the appropriate IES Lighting Measurement [LM] series guide.²⁻⁶)

5.3 Angular Position Sensing

The selection of appropriate angular position sensors is based on the uncertainty budget of the system and the resolution necessary to properly position the luminaire or instruments involved in its measurement. The selection of appropriate angular position sensors will assure an acceptable level of repeatability in individual measurements. Both the accuracy and the resolution of the angular position encoder are important to achieving the desired repeatability of the full test system.

There are a variety of position sensors currently available for use in photometric test equipment. Examples include: incremental optical encoder, single-turn absolute optical encoder, synchro or resolver with analog-to-digital (A/D) converter, inductosyn with A/D converter, high accuracy potentiometer with A/D converter, incremental magnetic encoder, and single-turn absolute magnetic encoder.

Of the available angular position sensing methods, optical encoders typically provide the highest resolution and accuracy. In recent years, magnetic angular encoders have become available with performance comparable to mid-range optical angular encoders. Caution should be exercised when considering inductosyns, synchros, resolvers, or

potentiometers. These technologies employ the use of both a sensor and an analog-to-digital (A/D) converter to measure angular position. It is important to consider both the sensor and the A/D converter's contribution in the uncertainty evaluation since the A/D converter is a critical contributor to the uncertainty of the angular sensing system. The use of potentiometers is generally discouraged as it provides the least direct method with lowest inherent accuracy.

It is important to note that the coupling of the angular position sensors to a photometer's mechanical structure and the rigidity and structural deflection properties of the photometer influence the "effective" resolution and accuracy of the photometric test equipment. Care shall be taken to ensure that structural deflection of mirror and luminaire mounts under the anticipated offset loads is less than the desired angular accuracy. Coupling of the encoders, when possible, should be direct to the final motion axis, via appropriate shaft couplings to avoid backlash (i.e. reversal error) in the measurement path. Care shall also be taken to ensure that the encoder-to-system coupling does not sacrifice planned accuracy due to installation misalignments such as a lack of shaft concentricity and/or shaft-to-shaft misalignment.

6.0 Electrical Measurements

6.1 General

Since modern lamp and ballast designs might produce non-sinusoidal wave shapes, alternating current (AC) instruments that measure root-mean-square (RMS) values are necessary. It is important that care shall be taken when selecting RMS measuring equipment. Not all instruments referred to as "RMS measuring instruments" accurately measure RMS values. Some instruments utilize a mathematical approach that assumes the waveform that is being measured is a perfect sine wave. This will lead to significant error when measuring wave shapes that differ from a pure sine wave. In addition, non-sinusoidal wave shapes can contain harmonic components that occur at high frequencies. To properly measure the contribution of these high frequencies, it will be necessary to select an instrument that is capable

of sampling at a frequency at least twice the frequency of the highest harmonic component. When selecting an instrument, the specifications for the instrument should be consulted to determine whether it is capable of properly measuring the electrical quantities for all wave shapes of interest.

Another consideration is the frequency response of the instrument. Many electronic ballasts operate at frequencies much higher than 60 Hz (typically on the order of 20 kHz). If the frequency of the quantity being measured is greater than the instrument's rated frequency response, significant errors in measurements can occur. When choosing an instrument for the laboratory, the manufacturer's specifications shall be consulted to verify that the instrument's frequency response is appropriate for the frequency of the waveform.

If an instrument is intended for use on circuits that have a direct current (DC) component in the waveform, additional care shall be taken. Not all instruments that measure AC waveforms are capable of accurately measuring the DC component of a given waveform. If the instrument will be used to measure waveforms that include a DC component, the manufacturer's specifications shall be consulted to verify that the instrument's DC response is of acceptable uncertainty.

6.2 Impedance Limitations

Another advantage of using digital instruments is that the impedance of the measurement circuit does not significantly affect the electrical quantities of the unit under test. The potential-measuring digital instruments are generally of high enough impedance that their effect on the unit under test can be ignored. Similarly, the current-measuring digital instruments are of low enough impedance that their effect on the unit under test may also be neglected.

For information pertaining to impedance concerns when using analog instruments the reader is referred to **Annex A**.

7.0 Optical Radiation Detectors and Measuring Instruments

7.1 General

The choice of instruments for measuring radiant energy includes both the selection of an appropriate optical radiation detector as well as the selection of the instruments used to measure the detector's output. Each combination of measuring equipment and detector poses unique concerns regarding instrument selection, calibration schedule, and levels of uncertainty. The manufacturer's specifications shall be consulted for all equipment involved to properly assess the uncertainty of the system.

7.2 Optical Radiation Detectors

Optical radiation detectors are devices that convert radiant energy to a measurable electrical quantity. The five most common types of detectors are photodiodes, charge-coupled devices (CCDs), photomultiplier tubes (PMTs), thermal detectors, and photoconductive detectors. *Photodiodes* are most commonly used for photometry and radiometry because of their excellent linearity and stability. Common types of photodiodes include silicon, germanium, and indium-gallium-arsenide. *Silicon photodiodes* are the most common detectors used in photometers. *CCDs* are photosensitive devices that convert local illuminance to a proportional electrical charge. This property makes them ideal for capturing the distribution of illuminance across their photosensitive area for applications such as luminance imaging and measurement of spectrally separated light (for example in a CCD array spectrometer). *PMTs* are vacuum tubes that couple a photosensitive element with high amplification. This combination makes them very sensitive detectors of ultraviolet, visible, and infrared radiation. *Thermal detectors and photoconductive detectors* are commonly used for measurements in the infrared region of the spectrum. Common types of thermal detectors include thermopiles, bolometers and pyroelectric detectors. Common types of photoconductive detectors include cadmium sulfide and cadmium selenide detectors. More information on detectors can be found in ANSI/IES LM-58-20.⁷

7.3 Amplifiers and Logging/Display Devices

Regardless of the type of detector that is selected, additional instruments are required to condition and measure the detector's output. These additional instruments can include circuitry for amplification of the signal, analog-to-digital converters for converting the analog electrical values to digital values, and computers or logging instruments for logging and display of the measured quantity. These types of instruments are available for purchase from a variety of sources. In many cases, the manufacturer of the detector can recommend compatible instruments for the logging and display of measured values.

7.4 Electrical Filtering

Luminaires and lamps that use magnetic ballasts can exhibit a phenomenon called flicker. Flicker is a periodic variation in the luminous output of a light source. In HID and fluorescent lamps, the flicker is caused by the arc in the lamp being initiated at alternate electrodes during positive and negative half-cycles of the supply voltage. In such lamps and luminaires, the luminous output will vary at twice the line frequency ($2 \times 60 \text{ Hz} = 120 \text{ Hz}$ in North America). If the detector used to measure the radiant energy from this type of source has a fast-enough response time, the output will follow this variation. If the source is measured directly and asynchronously relative to line frequency, erratic readings may result.

Luminaires and lamps that use LEDs as light sources can also be subject to flicker. In this case, it is a cyclical change in drive current that causes the flicker. This type of flicker can be particularly difficult to measure since it does not necessarily coincide with the supply line frequency. In addition, flicker in some LED sources and luminaires can be very dramatic, especially when AC waveforms or unfiltered pulse width modulation is used in the control circuitry. In these cases, the light output of the LED can range from the peak light output to no light output during the flicker cycle, requiring knowledge of the measuring instruments' limitations as well as the peculiarities of the test subject in order to ensure an accurate measurement.

With typical digital voltmeters, the signal is measured by integrating the voltage or current over a specific period

(integration time). Some instruments allow the user to select the integration time. By choosing the integrating time to be a multiple of the period of the line frequency, the effect of flicker for either line frequency can be removed. This method is often called *common mode noise rejection*. Although the flicker appears at twice the frequency of the line frequency, it is better to use a minimum of one complete cycle (not half cycle) of the power line frequency. This is recommended because the waveform of some discharge lamps may not be the same for the positive and negative phases of the line cycle. With this method, the amplifier response can be kept fast, and measurement can be repeated quickly without time-constant errors.

If the data acquisition is not of the integration type, the flicker component of the signal shall be removed either on the input analog signal or by many samplings at a fast rate and integrating (averaging) the readings over a power line cycle. If the flicker is to be removed on the analog signal, a low-pass filter with a sufficiently long time constant (several times the period of the AC power line) needs to be used. Care shall be taken since the longer time constant of the circuit might produce errors when data is acquired rapidly. This type of error is specifically prominent when the photometer signal is changing rapidly throughout the testing process (e.g., in the case of goniophotometry).

Another solution is to use a twice line-frequency notch filter. In this instance, care shall be taken that the data acquisition does not occur at or near the twice line-frequency rate. Data acquisition near this sample rate will cause the signal itself to be attenuated. Various other methods can be used depending on the construction of the system. In all cases, a thorough understanding of the cyclical nature of the light source is necessary to select the appropriate method of electrical filtering.

7.5 Linearity of Optical Radiation Measurement

The response of the detector-instrument system shall have linearity acceptable for the purposes of the measurement. This means that increases of energy incident on the detector will produce proportional changes in instrument indications. The linearity of the system is a factor in the overall instrument uncertainty.

The linearity of a radiometer or photometer depends not only on the linearity of the detector itself, but also on the electronics used to extract and amplify the detector output signal. An appropriate electronic circuit shall be used in conjunction with the detector to ensure linear operation. For example, a current-to-voltage trans-impedance amplifier is commonly used for silicon photodiodes. This type of electronic circuit is designed to keep the input impedance as low as possible (causing the voltage across the photodiode to be nearly zero).

A rough check of the linearity of the system (detector and electronic circuitry) can be performed on a photometric bench using the inverse-square law to verify that known changes in measurement distance produce appropriate changes in measured illuminance. Precise measurement of linearity, however, is not a simple task. The details of linearity measurements are out of the scope of this document. (Refer to ISO/CIE Publication 19476 and other references for further information on linearity requirements and linearity measurement.⁸⁻¹¹)

7.6 Cosine Response

When illuminance is measured where light is incident from a large range of incident angles, it is important that the photometer head (detector plus other components) responds according to a cosine curve with change in incidence angle. This "cosine response" is commonly achieved by the addition of a diffuser as the front surface of a detector. (For information regarding the evaluation of cosine-response error and correction methods, see ISO/CIE Publication 19476.¹¹)

7.7 Photopic Response

The spectral responsivity of detectors varies depending upon their physical construction. For measurement of photometric quantities (luminous intensity, illuminance, luminance, luminous flux), the photometer head shall have its relative spectral responsivity matched to the standardized human eye response—called the CIE spectral luminous efficiency function, $V(\lambda)$ —as closely as possible.¹² If the match is poor, there will be significant errors when measuring various types of lamps whose spectral distribution is dissimilar to the calibration source (standard incandescent lamp). The $V(\lambda)$ match is commonly achieved by use of colored glass filters that are tailored for each type of detector or each individual

detector. The degree of matching to $V(\lambda)$ is evaluated by a term called f_1 , defined in ISO/CIE 19476:2014.¹¹ (For more information regarding the fit of a detector to the $V(\lambda)$ curve, see **Annex C**.)

All colored filters experience some change in spectral response as the ambient temperature varies. In addition, the spectral responsivity of detectors also changes as the ambient temperature varies. For high-accuracy applications, the spectral response of the detector-filter combination should be evaluated at the temperature at which the detector will be used and should be maintained in a temperature-controlled environment. In all cases, the manufacturer's specifications should be consulted to determine whether the laboratory's temperature control is adequate to provide an acceptable level of uncertainty from the detector-filter combination.

8.0 Computer Interfacing of Photometric Equipment

There are many software and hardware combinations currently available that will interface with most instruments. In some situations, the computer, its hardware, and specially designed interfacing equipment can act as a standalone data collecting system. In all cases, the specifications for the interfacing equipment should be consulted to ensure that the interface will allow data transfer at a rate that is compatible with the requirements of the system.

9.0 Safety

9.1 General

The safety of the user should be considered always. In addition, the protection of the measuring equipment is an important consideration in the design of test procedures. General good housekeeping in the laboratory promotes safety and the pride in workmanship which leads to better overall results.

9.2 Electrical Shock

If possible, one side of the test circuit should be grounded. Grounding the test circuit is not only a safety precaution; it also removes the possibility of electrostatic errors in instrument readings.

Laboratory personnel should be particularly careful when working with equipment from which there is a chance of electrical shocks. Currents as small as 10 milliamperes can be harmful and, in some situations, fatal.

A shock can occur any time that a portion of the body becomes a parallel path to the electrical circuit. For this reason, it is important that electrical instruments and tools are properly grounded. It is also recommended that power cables be periodically checked to make sure that the insulation is not damaged or worn. The installation of a ground fault circuit interrupter (GFCI) can help prevent serious injury in the case of an electrical shock.

Electrical systems implemented in the laboratory should follow all local and national electrical code guidelines.

10.0 Additional Resources

Gottlieb IM. Basic Electronic Test Procedures, 2nd ed. New York: TAB Books, Inc.; 1985.

Levi L. Calibration and Recalibration of Photoelectric Meters to Correct for the Spectral Characteristics of the Measured Light. Illum Eng. 1960 Feb;55(2):99.

Walsh JWT. Photometry, 3rd ed. London: Constable and Co., Ltd; 1958.

Annex A – Analog Instrument Selection and Considerations

For a discussion of the corrections to compensate for the presence of test instruments in a lamp circuit, see the appropriate lamp photometry guides published by the IES.^{2,5,13}

A.1 Analog Instrument Selection

To minimize errors that can result from non-sinusoidal waveforms, it is recommended that voltmeters for measurement of lamp voltage of gaseous discharge lamps be selected from the following types (In order of preference from the standpoint of possible waveform errors): thermal, electrostatic, dynamometer and iron vane. (Note: Special requirements based on specific circumstances could produce a different order of preference.)

Thermal instruments are usually over-damped and have a fragile actuating system but retain accuracy over a wide range of frequencies. Electrostatic instruments are unstable on zero and require frequent calibration but should be used when discharge lamp currents are in the glow-discharge region. The dynamometer principle is the only one that can readily be applied to wattmeters. Iron vane instruments are the most rugged of all types; however, their accuracy is not equal to other types.

If measurements are to be made using standard reactors, wattmeters should be properly scaled to accurately read the parameters of the 7.5% power factor required for reactor setup. Wattmeters that are capable of measuring in this range of power factor are often called "low power factor" wattmeters.

A.2 Extending the Range

Instruments may be single-range or multi-range. In both cases, it is critical that care be taken to ensure that the readings are in the upper portion of the scale of the instrument.

It might be impractical to supply individual ac ammeters, voltmeters and wattmeters for all the ranges required in the laboratory. In the case where there is no DC component present, the use of current transformers can extend the current range, and the use of potential transformers can extend the voltage range. Caution should be used with range modifying transformers since they may seriously degrade the instrument's accuracy with respect to the frequency response rating.

When transformers are used to measure voltage or current, knowledge of the actual ratio of the transformers is necessary for an accurate measurement. For the measurement of power, the phase angles of the

transformers also need to be known. Once the phase angles of the transformers are known, corrections for the differences in phase relation of current and voltage within the wattmeter can be made.

A.3 Current Transformers

When measuring ac currents of 25 amperes or more, a current transformer with suitable primary ranges for use with a 5-ampere ammeter is generally preferable to a self-contained ammeter. This will keep heavy-current leads away from the instrument and isolate the measuring instrument from the possible high voltage of the supply circuit.

One multi-range current transformer with one ammeter can be used to replace several separate ac ammeters. The transformer should be tested to determine the ratio for its various ranges with its ammeter as a burden if the highest accuracy is desired. The primary of a current transformer should never be energized unless there is an ammeter or some other low resistance burden, such as a shorting link, across the secondary. In other words, the secondary should never be opened while the primary is energized. Otherwise, the secondary winding might have sufficiently high voltage induced in it to endanger insulation or the safety of personnel.

There is also the possibility that the magnetic condition of the core will be changed and alter the ratio and phase angle until the transformer is thoroughly demagnetized. This condition might also occur if direct current flows through the windings. If the secondary is left open for some time with the primary energized, overheating due to the greatly increased iron losses in the core might damage the insulation.

A.4 Potential Transformers

A potential transformer with a voltmeter connected to the secondary should be used for measuring AC voltages more than a few hundred volts. As a safety precaution, the secondary (low voltage winding) should be grounded at one point. For the highest accuracy, the transformer should be tested (ratio and phase angle determined) with the burden of its circuit.

A.5 Impedance Limitations

Analog instruments used for the measurement of

electrical values in gaseous discharge lamp circuits need to meet certain special requirements. The error of any instrument should not exceed 1%, or as required by the application, after calibration.

The instrument(s) chosen for voltage and current measurements shall be of the true RMS type and shall have a specified accuracy and frequency response adequate to ensure that these requirements are met. The potential coils should not draw more than 3% of the rated lamp current. If the current drawn by the potential coils is less than 1%, no correction for potential coil loss is necessary. The potential element of a wattmeter should always be connected so that the instrument measures the potential loss in this element.

The impedance of the ammeter and wattmeter current coils should produce a drop of less than 2% of the rated lamp volts. If the potential drop in the series elements of the ammeter and wattmeter current coils are less than 0.5% of rated lamp volts, no correction or compensation for the voltage drop is necessary.

Annex B – Tolerance Interval vs. Acceptance Interval

In many of the IES LMs, required conditions are stated in terms of a tolerance interval, with specified upper and lower tolerance limits. To ensure that any given

parameter is within the specified tolerance interval, the applicable measurement uncertainty shall be considered by deriving the corresponding acceptance interval. The acceptance interval is defined as the tolerance interval reduced by the expanded uncertainty of measurement (at 95% confidence) at both limits of the tolerance interval. This relationship is illustrated graphically in **Figure B-1**. The measured value of any specified parameter shall lie within the acceptance interval derived from the corresponding tolerance interval and measurement uncertainty.

As an example, the specifications for a commonly available AC power analyzer provides the accuracy of the AC voltage measurement for frequencies ranging from 45 Hz to 66 Hz as:

$$Accuracy = 0.1\% \times Reading + 0.1\% \times Range$$

This accuracy is valid for three months. For a 12-month calibration accuracy, the 3-month accuracy is multiplied by 1.5. This accuracy is revalidated when a calibration laboratory performs a single-point calibration or verification of a range setting on the AC power analyzer.

Therefore, for a 120-volt, 60-Hz reading (*Reading* = 120 V) on a 150-volt range (*Range* = 150 V), the accuracy is 0.41 V for 12 months, which defines the half-width of a uniform or rectangular distribution. Converting the half-width of a uniform distribution to standard uncertainty is done by dividing the half-width by the square root of three. The standard uncertainty for this

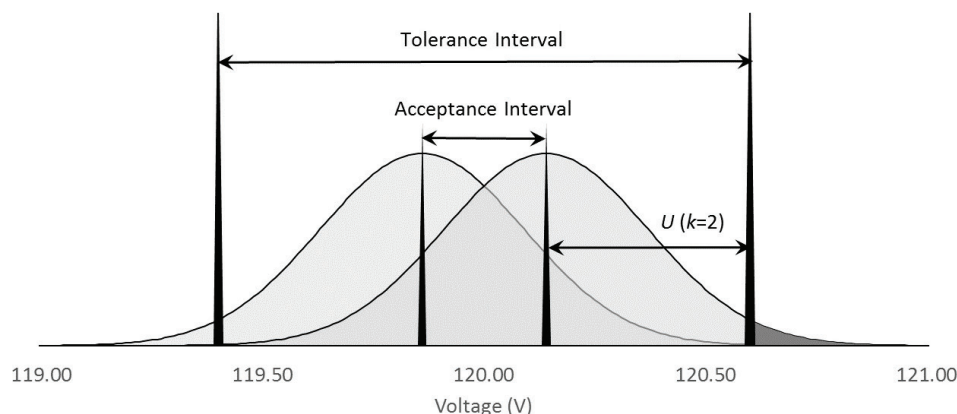


Figure B-1. Graphical relationship between tolerance interval, acceptance interval, and measurement uncertainty for a setting of 120 V AC. (© Illuminating Engineering Society)

AC power analyzer for AC voltage measurement is 0.23 V, or 0.19%. With a coverage factor of $k=2$, the expanded uncertainty or calibration uncertainty is 0.46 V, or 0.38%. For measuring 220 V, the expanded uncertainty is 0.90 V, or 0.41%.

To set the AC voltage to 120.00 V using this AC power analyzer, for which the tolerance interval is ± 0.5 %, the AC power analyzer should read between 119.86 V and 120.14 V (which defines the acceptance interval). Further information on the concept of acceptance interval is provided in ISO/IEC Guide 98-4.¹

Another example is the measurement of the ambient temperature, which shall be maintained at 25 °C with a tolerance interval of ± 1.2 °C. If the expanded uncertainty ($k=2$) of the thermometer is 0.2 °C, the reading of the thermometer shall be ± 1.0 °C from 25 °C, as shown in **Figure B-2**.

Annex C – Measurement Errors Due to the Deviation in the System Response from $V(\lambda)$

The percent error possible in a measurement for any detector is the maximum percent deviation from the $V(\lambda)$ curve. Knowledge of the detector's response when not accompanied by knowledge of the spectra of the source under test can lead to large errors even for a

detector with a small f_1' . Measurements of low pressure sodium (LPS) lamps or light emitting diodes (LED), which may have all their radiant energy in narrow frequency bands, are susceptible to this type of error.

In some cases, the unknown errors are not significant, and if the person making the measurement ignores them there are no serious consequences. In other situations, relative readings suffice, and absolute levels are not important. If, however, the objective is to obtain a light output measurement with a known level of uncertainty, the cited references should be consulted and the recommended procedures carried out.^{11,14,15,16}

C.1 Illuminance

Commercially available photometric detectors only approximate the standard $V(\lambda)$ response due to limitations in the manufacturing process. Generally, the closer the "fit" to $V(\lambda)$, the more expensive the detector. A numerical term used to express the accuracy of fit is f_1' . It is important to be aware that this term provides an indication of the average closeness of the detector response to the ideal curve (i.e., its overall conformance to the standard); f_1' does not provide information about the magnitude of deviation of the response from $V(\lambda)$ at specific wavelengths. To determine the magnitude of error of a measurement due to detector spectral mismatch, the following information is required:

- The relative spectral response of the detector
- The spectral distribution of the source being measured

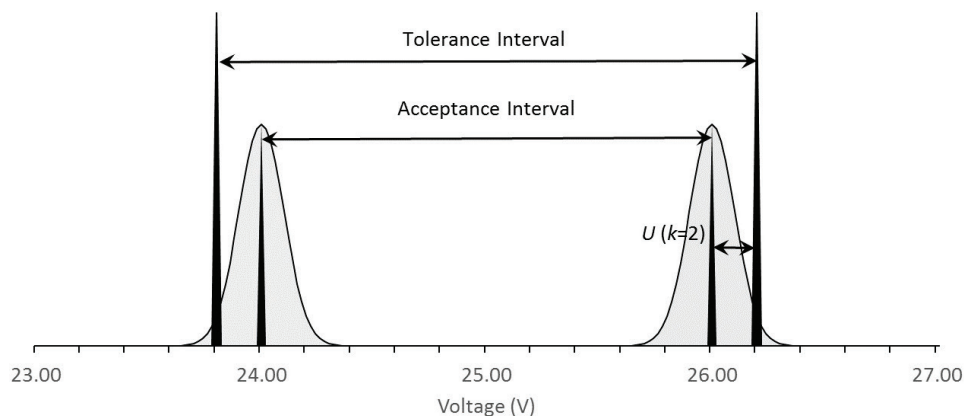


Figure B-2. Graphical relationship between tolerance interval, acceptance interval, and measurement uncertainty for ambient temperature measurement. (© Illuminating Engineering Society)

- The spectral distribution of the source used to calibrate the detector

The relative spectral response of a detector is often available from the manufacturer and it can be measured at national metrology institutes such as the U.S. National Institute of Standards and Technology (NIST), or at one of several commercial laboratories. Provided that the necessary equipment and the expertise are available, the detector response can be measured. The spectral distributions of the source under test and the calibration source are determined using a spectroradiometer. A correction can then be calculated and applied.

C.2 Total Luminous Flux

For the measurement of total luminous flux, the detector is positioned at the exit port of an integrating sphere. Integrated, diffuse radiation from a light source is then measured by the detector. Since the reflectance of the sphere wall varies with wavelength, the relative response of the detector-sphere combination will differ from the response of the detector alone. As in the case with the illuminance measurements (see **Annex B, Section B.1**), the inaccuracies introduced into the measurement depend on the difference in spectral power distribution of the source under test from that of the calibration source. For example, if the test source and the calibration source are both incandescent lamps operating at the same correlated color temperature, the error in the calibration is cancelled out by the same error in the measurement of the test source. On the other hand, if the calibration source is an incandescent source and the test source is a discharge lamp, the error may be significant. As in the case of the detector used directly for illuminance measurements, the magnitude of error can be determined and corrections applied using the same method described in **Section B.1**; however, the effects of the sphere need to be taken into account.

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Process for Change to an ANSI/IES Standard under Continuous Maintenance

This standard is maintained under continuous maintenance procedures, for which IES has an established and documented program for regular publication of addenda or revisions, including procedures for timely, documented, consensus action on requests for change to any part of the standard. Committee consideration will be given to proposed changes by June 30 of any given year for proposed changes received by the IES Director of Standards no later than December 31 of the previous year.

Submittal Format

Proposed changes must be submitted to the IES Director of Standards in the announced published format. However, changes may be accepted in an earlier published format, if the differences are immaterial to the proposed change submittal. If the Director of Standards concludes that a current form must be utilized, the proposer may be given up to 20 additional days to resubmit the proposed changes in the current format.

Specific changes in the text or values are required and must be substantiated. Any change proposals that do not meet these requirements will be returned to the proposer. Supplemental background documents to support changes submitted may be included.

Submission to the Committee Chair

The Director of Standards shall forward proposed changes received on appropriate forms to the committee chair for assigning to committee members (responders) to develop responses to submitters of proposed changes.

Review and Clarification

Responders shall review proposals and should contact the proposer if necessary for clarification.

Response Recommendation

Designated responders shall draft a recommended committee response, including any recommended changes to the standard. The 'responders' recommended responses shall be submitted to the committee chair in electronic form usable by Society Staff, including any recommended change to the standard in response to proposals received.

Options for Committee response are limited to:

- a) Proposed change accepted for public review without modification
- b) Proposed change accepted for public review with modification
- c) Proposed change accepted for further study
- d) Proposed change rejected

The responders shall provide reasons for any recommendation other than option (a) above.

The designated responders shall not recommend option (c) unless the further study can be completed by October 1 of that year, and providing the Committee can then vote for option (a), (b), or (d) no later than November 15 of that year.

Editing

The Committee chair or his or her designee shall edit the draft responses and circulate the edited drafts to the committee for review.

Form for Proposing Change to an ANSI/IES Standard under Continuous Maintenance

NOTE: Use a separate form for each comment. Submit to the Director of Standards, IES, 120 Wall Street, 17th Floor, New York, NY 10005-4001. Email: standards@ies.org. Fax: 212-248-5017.

1. Submitter: _____
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Telephone: _____
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I hereby grant the Illuminating Engineering Society (IES) the non-exclusive royalty rights, including non-exclusive rights in copyright, in my proposals. I understand that I acquire no rights in publication of the standard in which my proposals in this, or other analogous, form are used. I hereby attest that I have the authority and am empowered to grant this copyright release.

Submitter's signature: _____ Date: _____

2. Title of publications and year published _____

3. Clause (section), sub-clause or paragraph number; and page number: _____

4. My proposal (check one):

- ☐ Change to read as follows
- ☐ Delete and substitute as follows
- ☐ Add new text as follows
- ☐ Delete without substitution

Use underscore to show material to be added (added) and strikethrough for material to be deleted (~~deleted~~). Use additional pages if needed.

5. Proposed change:

6. Reason and substantiation:

Select as applicable:

- ☐ Additional pages are attached. Number of additional pages: _____
- ☐ Attachments or referenced materials cited in this proposal accompany this proposed change.

Please verify that all attachments and references are relevant, current, and clearly labeled to avoid processing and review delays. Please list your attachments here:



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